BIRZEIT UNIVERSITY

Department of Computer Science

First Semester 2025/2026

(COMP338) - (AI)

Course Project 2

Due date: Jan 23, 2026

Project Title:

Classify News Headlines into Categories (Text) by using Decision Tree and Logistic Regression

Objective:

In this project, students will build and evaluate machine learning models that automatically classify news headlines into predefined categories (e.g., World, Sports, Business, and Science/Technology). The project focuses on applying Decision Tree and Logistic Regression algorithms to a real-world text classification problem.
Students will gain hands-on experience with text preprocessing, feature extraction, model training, evaluation, and comparison, while developing an understanding of how different algorithms perform on the same dataset.

By completing this project, students will be able to:

- Understand the end-to-end machine learning pipeline for text classification
- Preprocess and clean textual data (news headlines)
- Convert text into numerical features using techniques such as TF-IDF Vectorizer
- Train and evaluate Decision Tree and Logistic Regression models
- Compare the performance of Logistic Regression and Decision Tree models by using: Accuracy, Precision, Recall and F1-score.
- Interpret results and discuss strengths and weaknesses of each model

The Dataset:

The datasets containing:
Feature (Input):
Text: a string feature contains short news title or headline
Label (Output):
label: a classification label, with possible values including World (0), Sports (1), Business (2), Sci/Tech (3).

Link to the dataset:
https://huggingface.co/datasets/wangrongsheng/ag_news

What to Submit:

Each team is expected to submit a working implementation of the both algorithms and a Report.
Implementation and Experimental work should include:

- Preprocess and clean textual data (news headlines)
- Convert text into numerical features using techniques such as TF-IDF Vectorizer
- Train and evaluate Decision Tree and Logistic Regression models
- Compare the performance of Logistic Regression and Decision Tree models by calculating Accuracy, Precision, Recall and F1-score.
- Generate and display graph(s) for Logistic Regression and Decision Tree models
- Interpret results and discuss strengths and weaknesses of each model

Your report should include:
● Decision Tree and Logistic Regression algorithms
● overview on the dataset
● If you using TF-IDF Vectorizer or any other techniques, explain the chosen method and parameters
● explain Accuracy, Precision, Recall and F1-score
● present your results
● discuss the difference in performance measurements between Logistic Regression and Decision Tree models
● explain your source code
● present the decision tree.

Grading Criteria:

● Grades are distributed as follows: Implementation: 33% Report: 33% Discussion: 33%
● For the project: - proper implementation of the algorithms - calculation of Accuracy, Precision, Recall and F1-score - generating the decision tree(s) - implementing the tree

● For the report: - explanation of necessary algorithms and evaluation metrics - overview on the dataset - explanation of the code - discussion of the experimental work, specially the difference in evaluation metrics between logistic Regression and Decision Tree models- final results and conclusion
● Discussion: You will be asked several questions some to verify your contribution to the project implementation and some to grade your understanding of the material itself Each student will be graded separately, so team members may have different scores.

Tools and Technologies:

- Programming Language: Java, Python, R, Octave
- Libraries: Weka or NumPy, Pandas, Matplotlib (for plotting)

Important Rules:

- Late submissions will not be graded reply before midnight of 23th of January
- The project is a teamwork, each team consists of two students
- Team members can be from different sections
- There will be a project discussion and it will have 33% of the project scores
- Plagiarism is not tolerated and can lead to failing the whole course